



USN

NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25MCS102 – ADVANCED COMPUTER NETWORKS

(For MCS)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Module**.

Module – I

	Marks	BT*	CO*	PO*
1. a) Demonstrate the different transmission strategies from IPv4 to IPv6	10	3	1	2
b) A packet has arrived in which the offset value is 100, the value of HLEN is 5, and the value of the total length field is 100. What are the numbers of the first byte and the last byte?	5	3	1	2
c) Discuss the four fundamental characteristics of data communication system.	5	2	1	1
2. a) Explain IPv6 Header format.	10	2	1	1
b) Discuss fragmentation and demonstrate the significance of offset field and flags in IPv4 with example.	10	2	1	1

Module – II

3. a) Explain TCP Segment format.	10	2	2	2
b) Develop process to process delivery and illustrate socket address, multiplexing and demultiplexing by transport layer.	10	2	2	2
4. a) Illustrate UDP operations in detail.	10	2	2	2
b) Explain leaky bucket and token bucket algorithm.	10	2	2	1

Module – III

5. a) Explain HTTP streaming and illustrate the interaction between client and server for HTTP streaming.	10	2	3	2
b) Illustrate three network-level approaches to support multimedia.	10	2	3	2
6. a) Construct RTP Packet Header Fields. Identify its importance and describe in detail.	10	3	3	1
b) Demonstrate CDN operations. Identify the importance of those operations and describe in detail.	10	3	3	1

Module – IV

7. a) Demonstrate 3G cellular network architecture. Identify and describe in detail the components.	10	3	4	2
b) Illustrate some of the key fields in the agent advertisement message.	10	2	4	1
8. a) Explain the 802.11 MAC Protocol.	10	2	4	2
b) Illustrate the two important innovations of the 4G Long-Term Evolution (LTE) standard over 3G systems. Compare their features.	5	3	4	2
c) Illustrate the Initial elements of mobile network architecture.	5	2	4	1

Module – V

9. a) Identify the components and demonstrate Juniper Network SDN Framework.	10	3	5	2
b) Identify the components and explain hardware and software components of sensor nodes.	10	2	5	1
10. a) Demonstrate wireless sensor network application in Home control systems.	10	3	5	2
b) Explain the importance of SDN.	10	2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
